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(54) **RETINAL PROJECTION DISPLAY SYSTEM**

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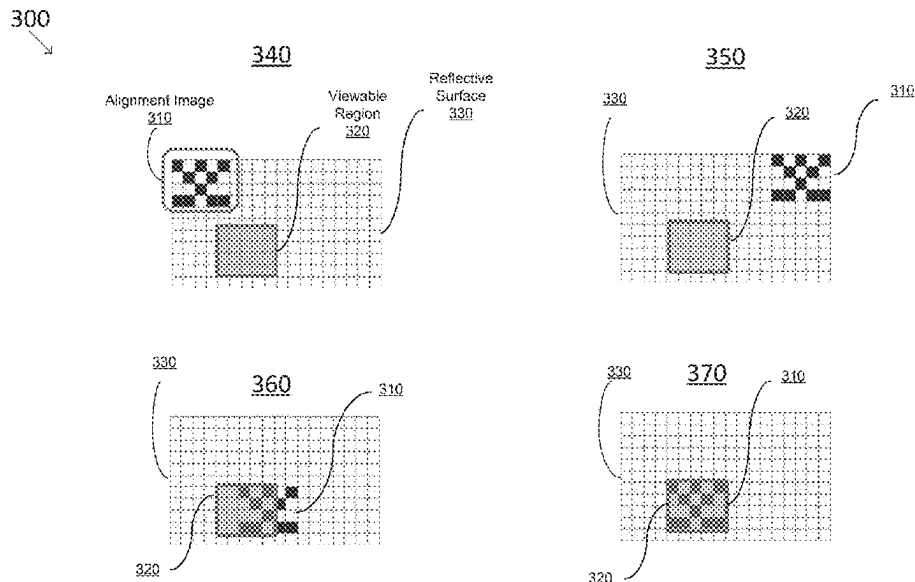
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(57) **ABSTRACT**

A retinal projection display system includes a light source for projecting an image, a scanning mirror having a field of view larger than the image, and a reflective surface on which the image is projected, wherein the reflective surface is larger than the image. The scanning mirror projects the image onto a viewable region of the reflective surface such that the image is projected into a retina of a user.

18 Claims, 10 Drawing Sheets



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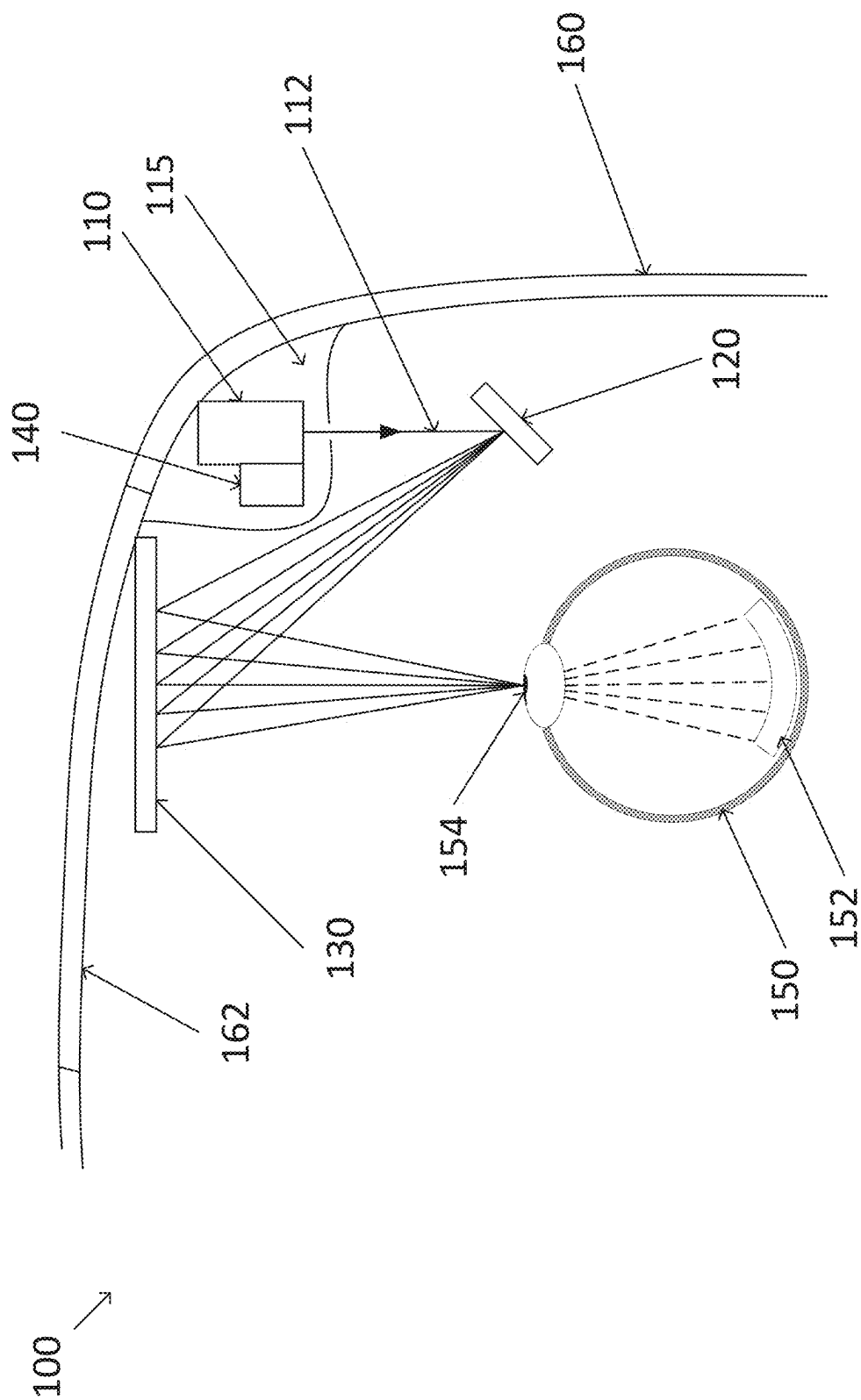


FIG. 1

200 ↗

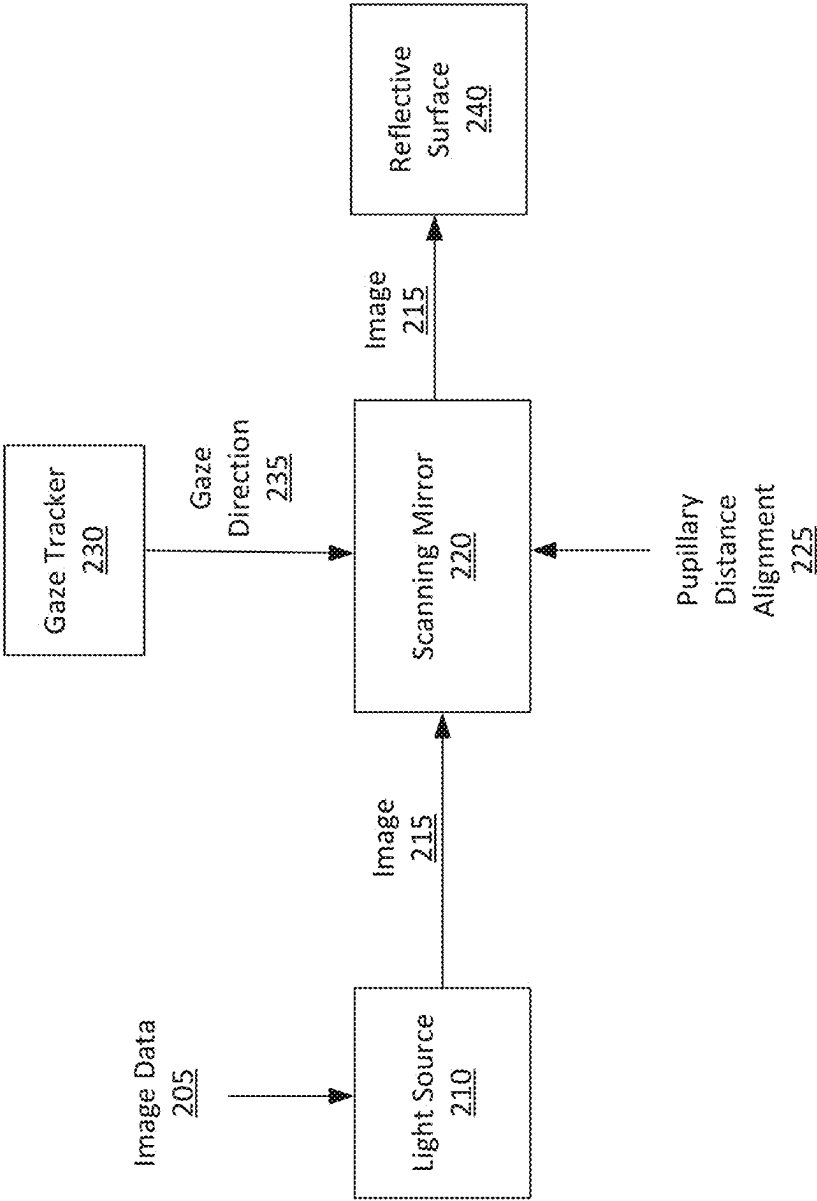
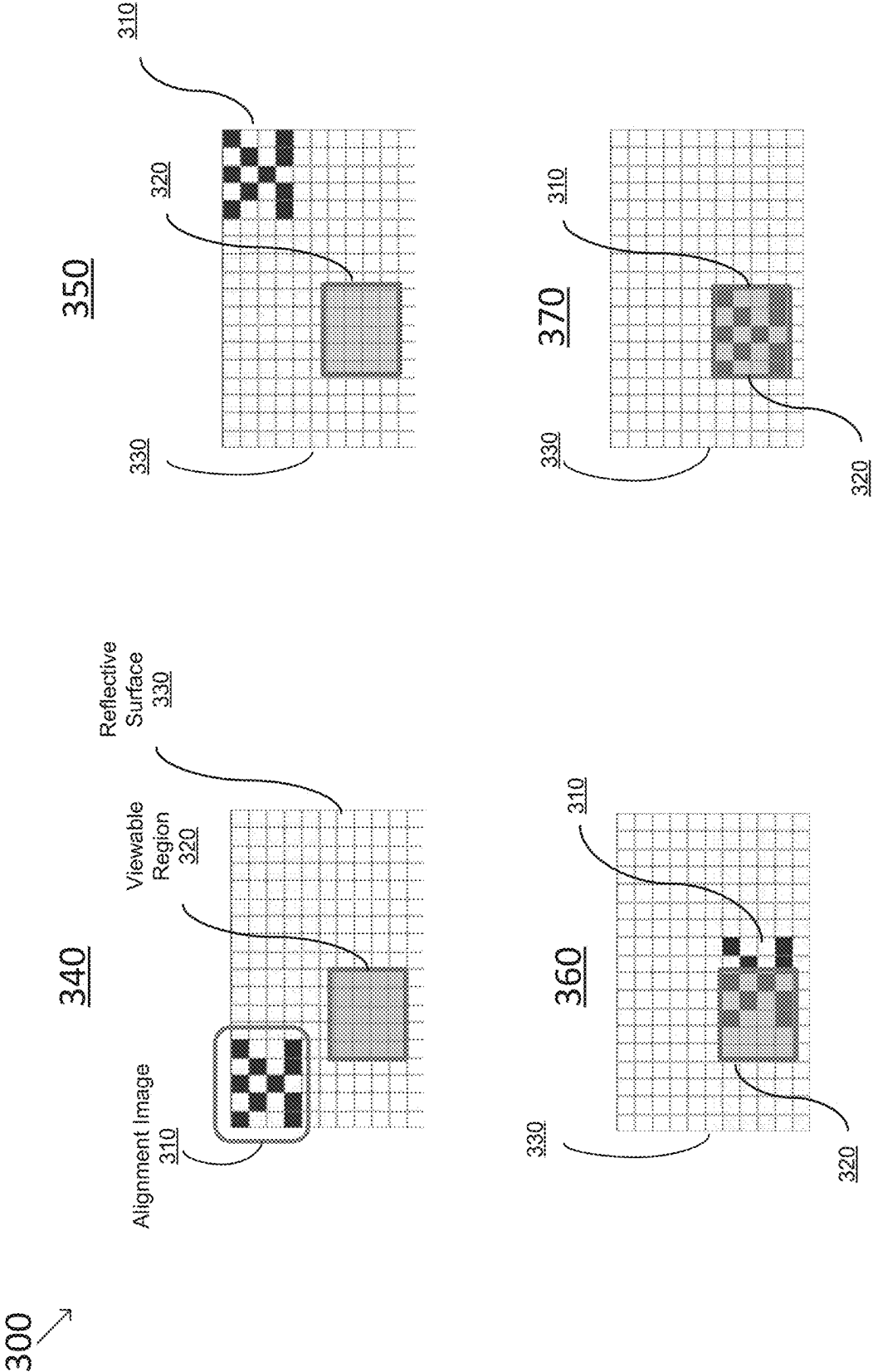
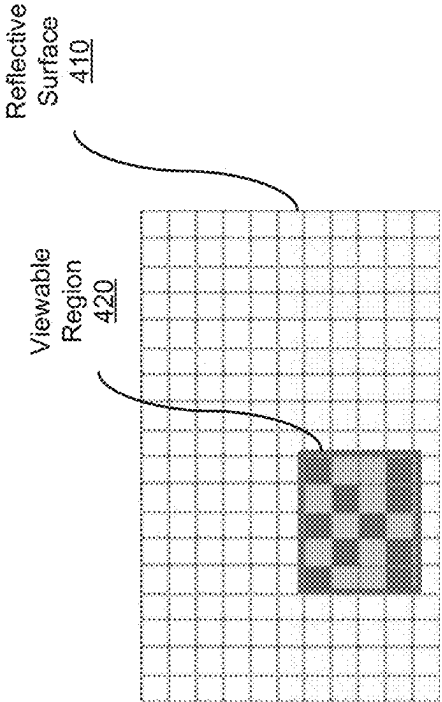


FIG. 2



400 ↗

430



440

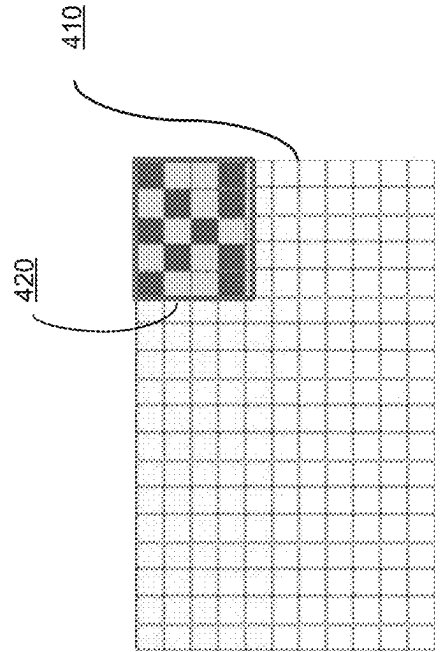


FIG. 4

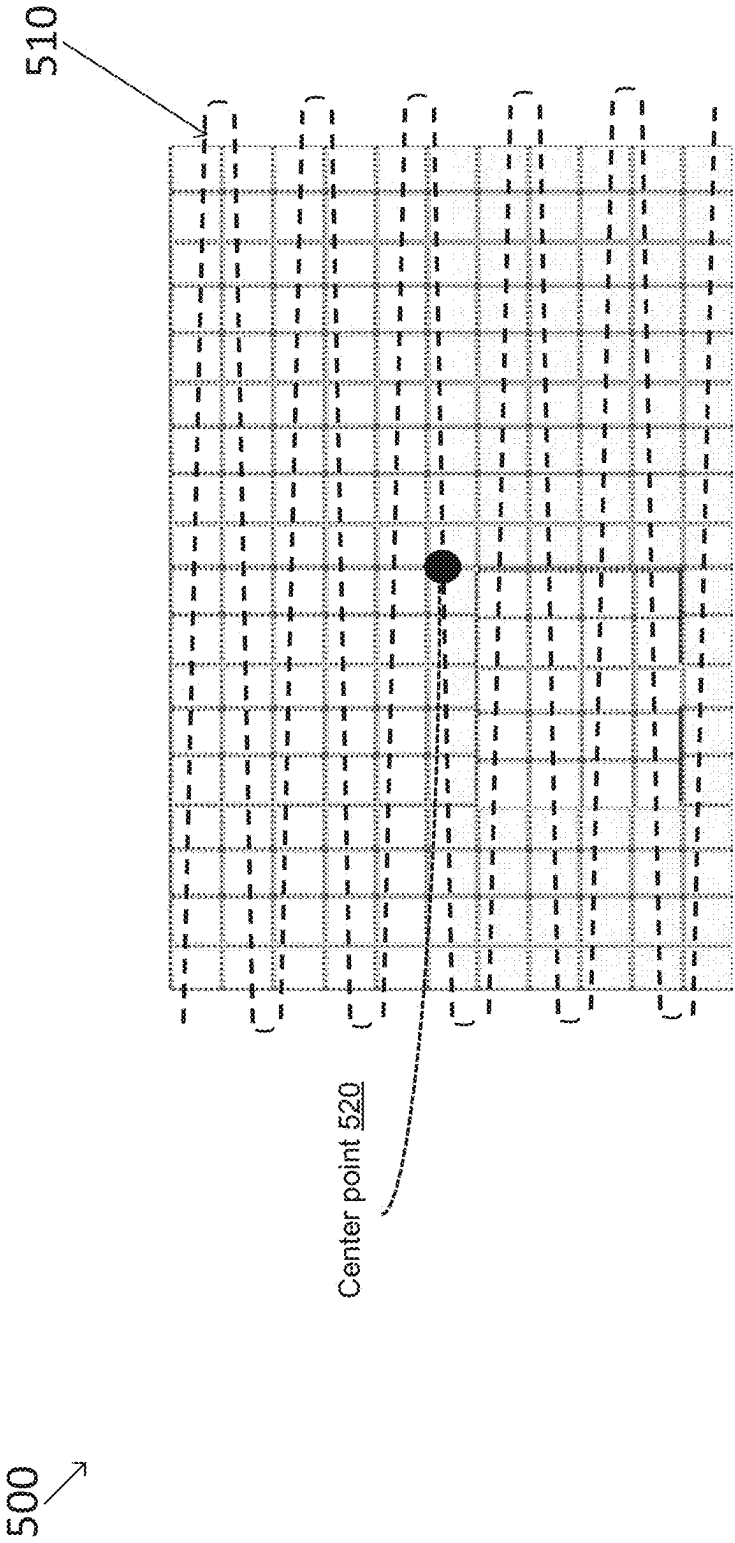


FIG. 5A

500

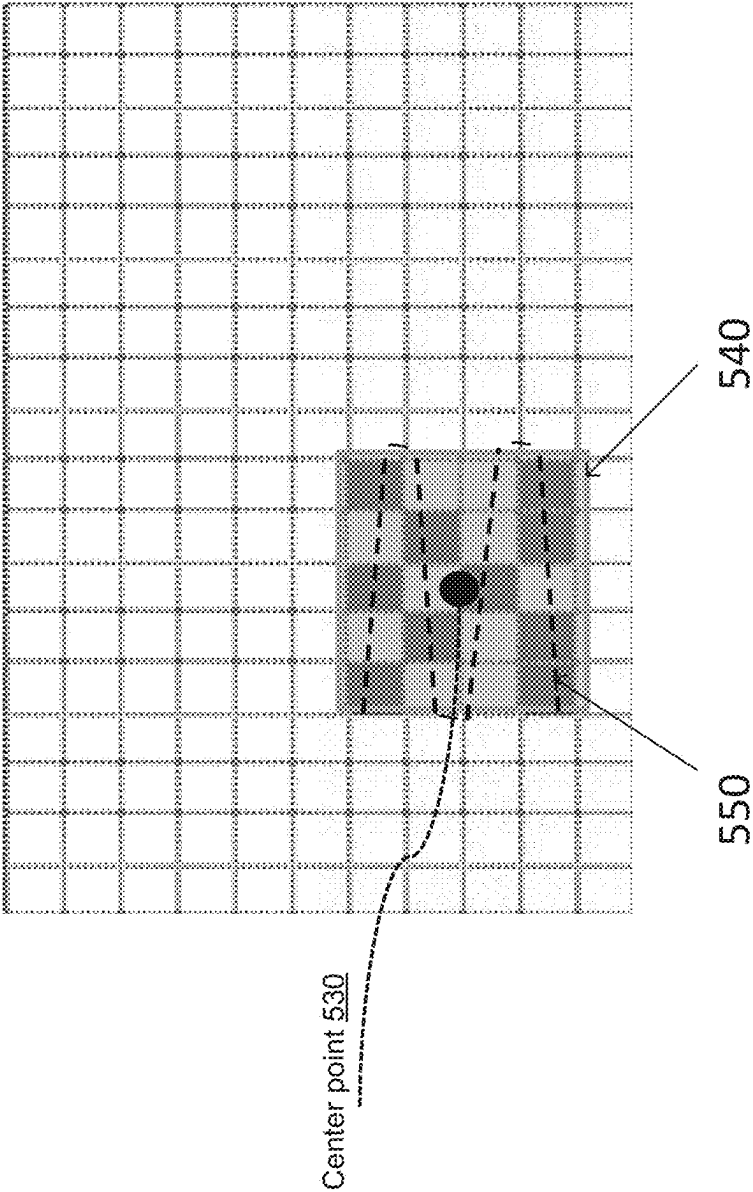


FIG. 5B

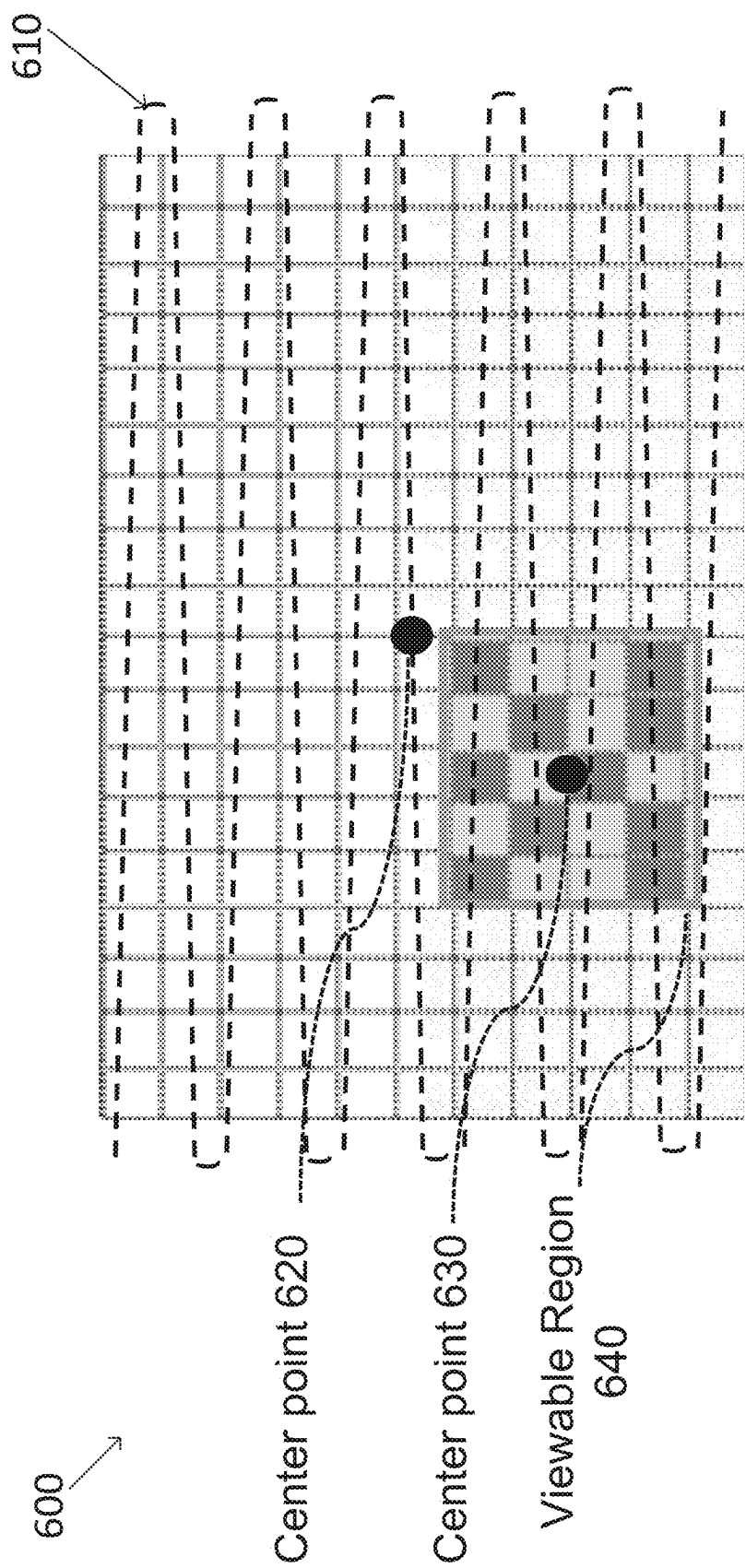
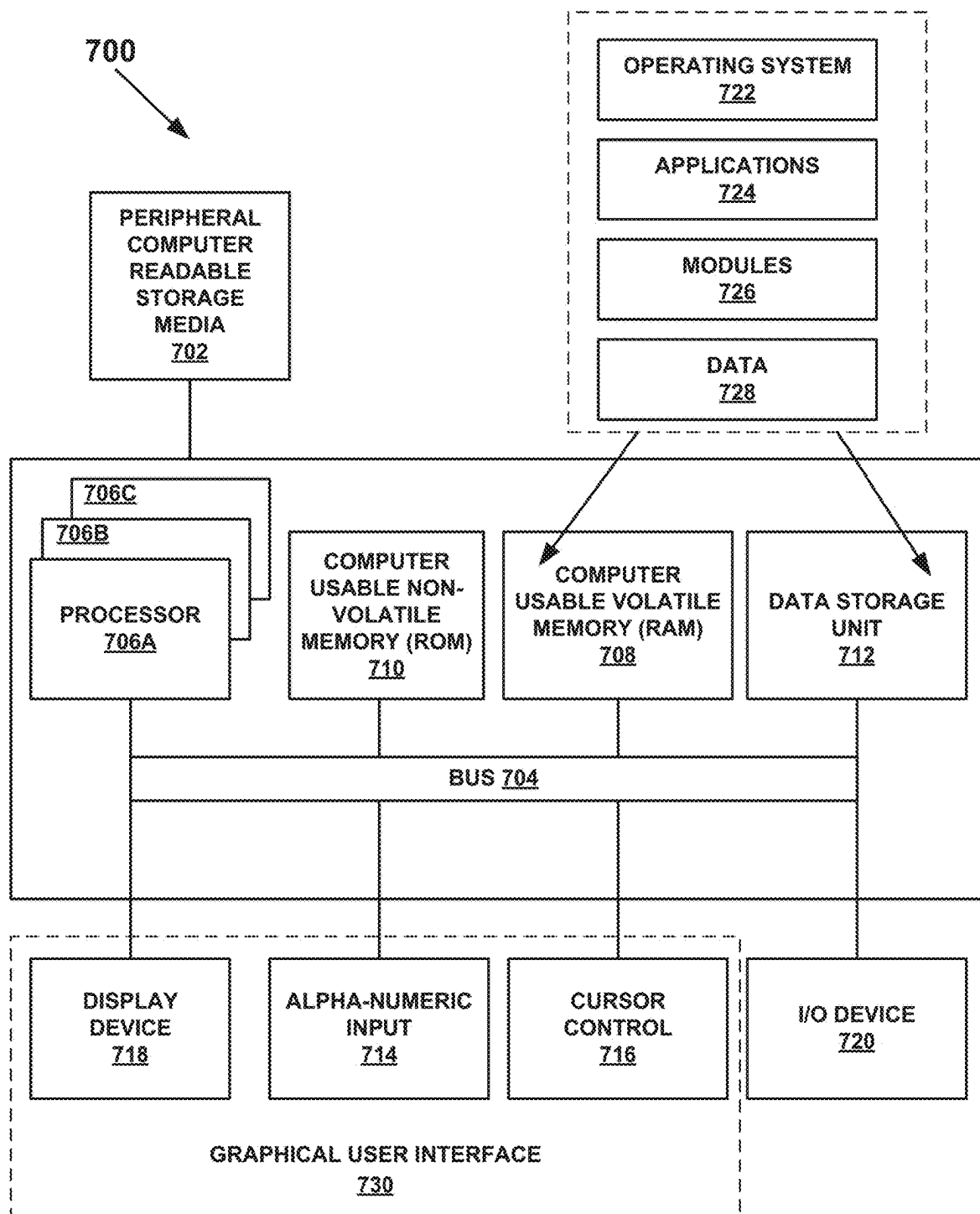


FIG. 6

**FIG. 7**

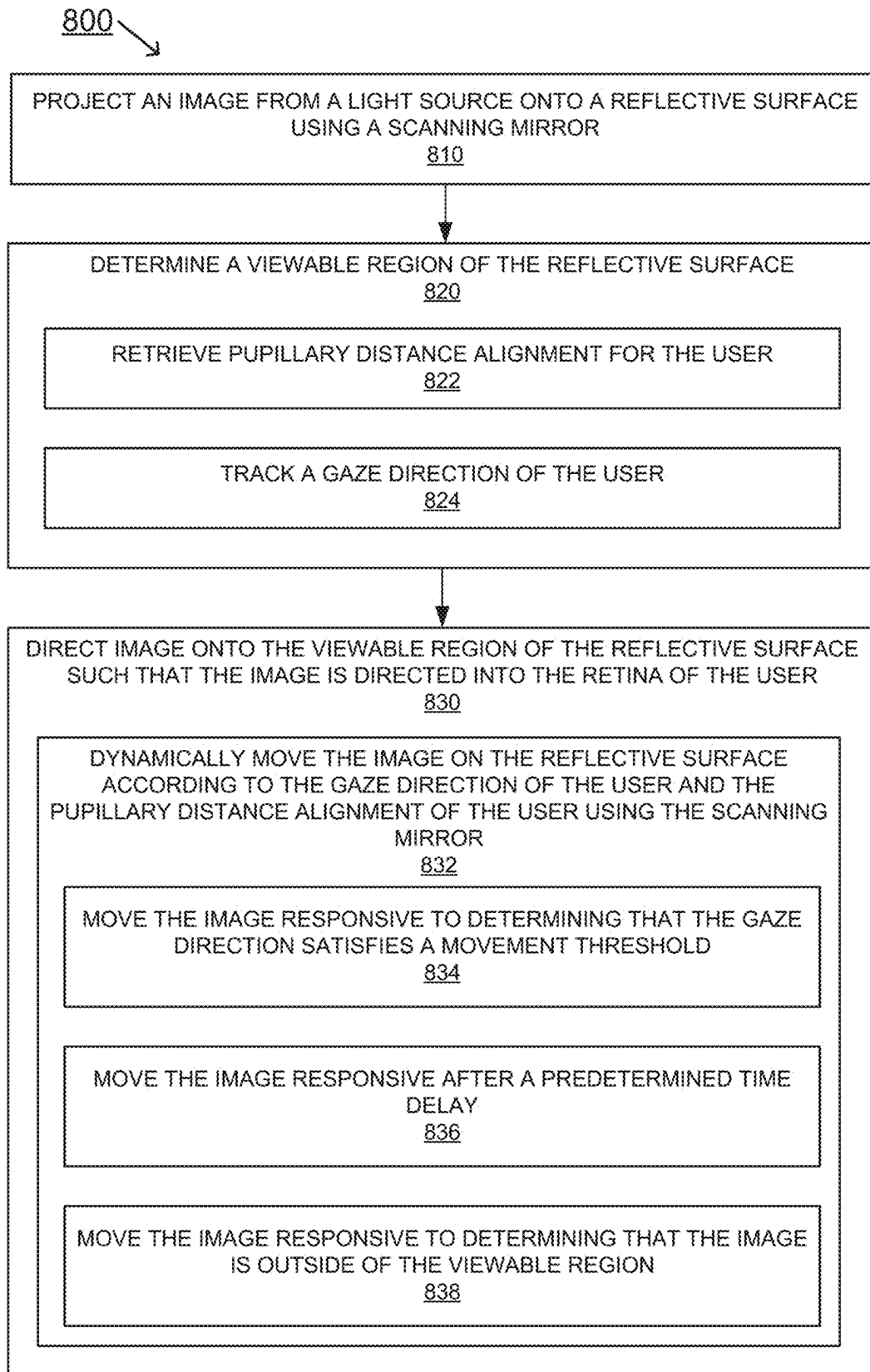


FIG. 8

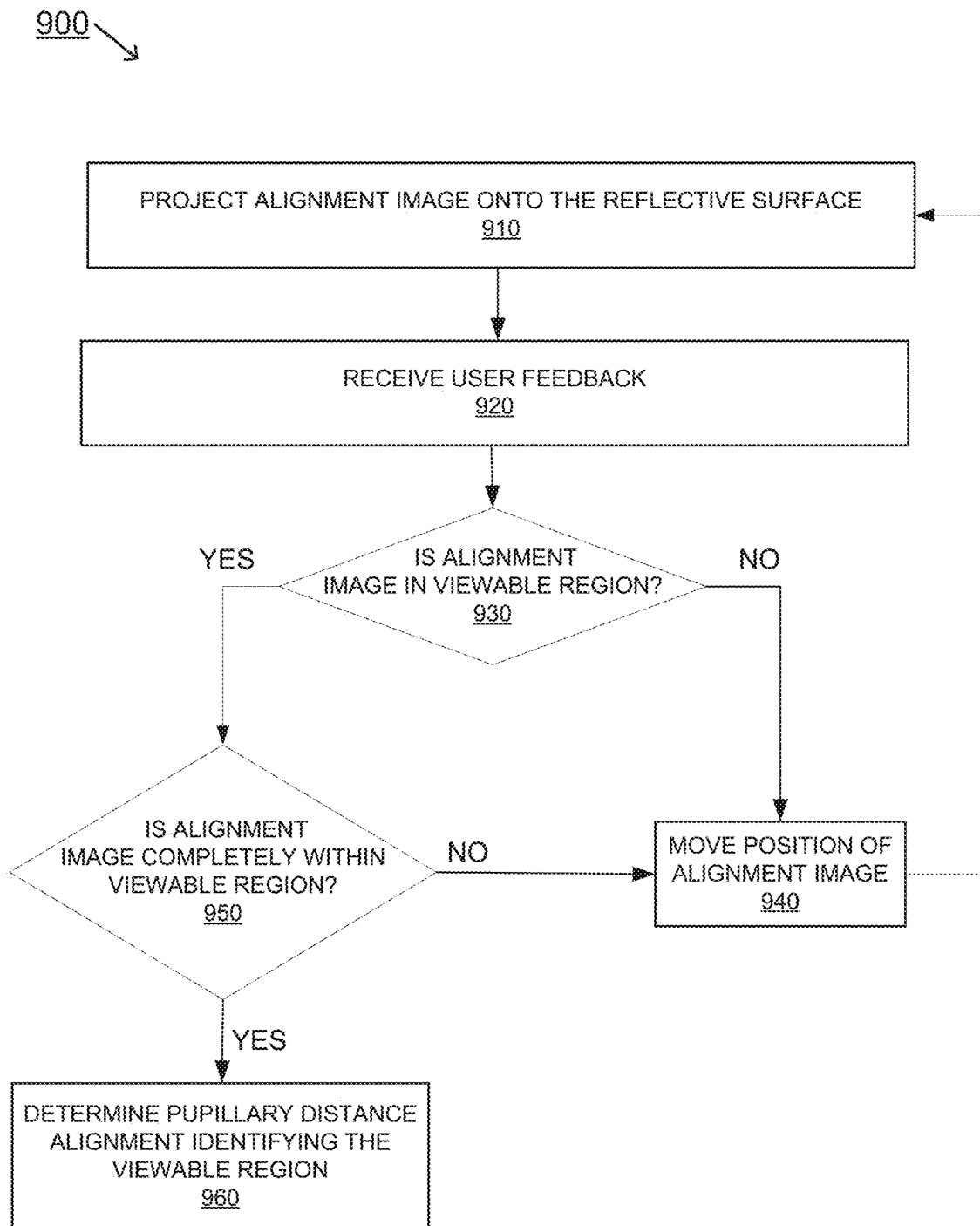


FIG. 9

RETINAL PROJECTION DISPLAY SYSTEM

RELATED APPLICATION

This application claims priority to and the benefit of U.S. Provisional Patent Application 63/260,485, filed on Aug. 20, 2021, entitled "ADAPTIVE EYE-BOX IN AR SMART GLASSES," by Heshmati, et al., and assigned to the assignee of the present application, which is incorporated herein by reference in its entirety.

BACKGROUND

Retinal projection displays (RPDs), also referred to as virtual retinal displays (VRD), are used to project images through the pupil of an eye directly onto the retina. The image rendering is performed fast enough such that the human eye perceives a continuous video stream of images. As the area through which the images are projected through the pupil and onto the retina, also referred to as the "eye box," is small, it is essential to have precise alignment between the RPD and the eye to ensure that the image enters the eye. Furthermore, as the gaze direction of a user can change during usage of an RPD, thus changing the location of the eye box, it is necessary to account for the change in gaze direction during usage of the RPD.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are incorporated in and form a part of the Description of Embodiments, illustrate various non-limiting and non-exhaustive embodiments of the subject matter and, together with the Description of Embodiments, serve to explain principles of the subject matter discussed below. Unless specifically noted, the drawings referred to in this Brief Description of Drawings should be understood as not being drawn to scale and like reference numerals refer to like parts throughout the various figures unless otherwise specified.

FIG. 1 illustrates an example retinal projection display system, according to some embodiments.

FIG. 2 illustrates a functional block diagram of an example retinal projection display system, according to some embodiments.

FIG. 3 illustrates an example pupillary distance alignment operation by projecting an alignment image onto a reflective surface, according to some embodiments.

FIG. 4 illustrates an example gaze tracking operation used during projection of an image onto a reflective surface, according to some embodiments.

FIG. 5A illustrates an example scanning pattern over the full scanning range of the scanning mirror, according to an embodiment.

FIG. 5B illustrates an example scanning pattern where the scanning range of the scanning mirror is the size of the projected image.

FIG. 6 illustrates an example image rendering operation where the scanning range of the scanning mirror is larger than the size of the projected image, according to an embodiment.

FIG. 7 is a block diagram of an example electronic device upon which embodiments described herein may be implemented.

FIG. 8 illustrates an example process of retinal projection, according to some embodiments.

FIG. 9 illustrates an example process for determining a pupillary distance alignment, according to some embodiments.

DESCRIPTION OF EMBODIMENTS

The following Description of Embodiments is merely provided by way of example and not of limitation. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding background or in the following Description of Embodiments.

Reference will now be made in detail to various embodiments of the subject matter, examples of which are illustrated in the accompanying drawings. While various embodiments are discussed herein, it will be understood that they are not intended to limit to these embodiments. On the contrary, the presented embodiments are intended to cover alternatives, modifications and equivalents, which may be included within the spirit and scope the various embodiments as defined by the appended claims. Furthermore, in this Description of Embodiments, numerous specific details are set forth in order to provide a thorough understanding of embodiments of the present subject matter. However, embodiments may be practiced without these specific details. In other instances, well known methods, procedures, components, and circuits have not been described in detail as not to unnecessarily obscure aspects of the described embodiments.

Notation and Nomenclature

Some portions of the detailed descriptions which follow are presented in terms of procedures, logic blocks, processing and other symbolic representations of operations on data within an electrical device. These descriptions and representations are the means used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. In the present application, a procedure, logic block, process, or the like, is conceived to be one or more self-consistent procedures or instructions leading to a desired result. The procedures are those requiring physical manipulations of physical quantities. Usually, although not necessarily, these quantities take the form of acoustic (e.g., ultrasonic) signals capable of being transmitted and received by an electronic device and/or electrical or magnetic signals capable of being stored, transferred, combined, compared, and otherwise manipulated in an electrical device.

It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. Unless specifically stated otherwise as apparent from the following discussions, it is appreciated that throughout the description of embodiments, discussions utilizing terms such as "performing," "determining," "detecting," "directing," "calculating," "correcting," "providing," "receiving," "analyzing," "confirming," "displaying," "presenting," "using," "completing," "instructing," "comparing," "executing," "tracking," "moving," "retrieving," "projecting," "calibrating," or the like, refer to the actions and processes of an electronic device such as an electrical device.

Embodiments described herein may be discussed in the general context of processor-executable instructions residing on some form of non-transitory processor-readable medium, such as program modules, executed by one or more computers or other devices. Generally, program modules

include routines, programs, objects, components, data structures, etc., that perform particular tasks or implement particular abstract data types. The functionality of the program modules may be combined or distributed as desired in various embodiments.

In the figures, a single block may be described as performing a function or functions; however, in actual practice, the function or functions performed by that block may be performed in a single component or across multiple components, and/or may be performed using hardware, using software, or using a combination of hardware and software. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, logic, circuits, and steps have been described generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present disclosure. Also, the example ultrasonic sensing system and/or mobile electronic device described herein may include components other than those shown, including well-known components.

Various techniques described herein may be implemented in hardware, software, firmware, or any combination thereof, unless specifically described as being implemented in a specific manner. Any features described as modules or components may also be implemented together in an integrated logic device or separately as discrete but interoperable logic devices. If implemented in software, the techniques may be realized at least in part by a non-transitory processor-readable storage medium comprising instructions that, when executed, perform one or more of the methods described herein. The non-transitory processor-readable data storage medium may form part of a computer program product, which may include packaging materials.

The non-transitory processor-readable storage medium may comprise random access memory (RAM) such as synchronous dynamic random access memory (SDRAM), read only memory (ROM), non-volatile random access memory (NVRAM), electrically erasable programmable read-only memory (EEPROM), FLASH memory, other known storage media, and the like. The techniques additionally, or alternatively, may be realized at least in part by a processor-readable communication medium that carries or communicates code in the form of instructions or data structures and that can be accessed, read, and/or executed by a computer or other processor.

Various embodiments described herein may be executed by one or more processors, such as one or more motion processing units (MPUs), sensor processing units (SPUs), host processor(s) or core(s) thereof, digital signal processors (DSPs), general purpose microprocessors, application specific integrated circuits (ASICs), application specific instruction set processors (ASIPs), field programmable gate arrays (FPGAs), a programmable logic controller (PLC), a complex programmable logic device (CPLD), a discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein, or other equivalent integrated or discrete logic circuitry. The term "processor," as used herein may refer to any of the foregoing structures or any other structure suitable for implementation of the techniques described herein. As it employed in the subject specification, the term "processor" can refer to substantially any computing processing unit or

device comprising, but not limited to comprising, single-core processors; single-processors with software multithread execution capability; multi-core processors; multi-core processors with software multithread execution capability; multi-core processors with hardware multithread technology; parallel platforms; and parallel platforms with distributed shared memory. Moreover, processors can exploit nano-scale architectures such as, but not limited to, molecular and quantum-dot based transistors, switches and gates, in order to optimize space usage or enhance performance of user equipment. A processor may also be implemented as a combination of computing processing units.

In addition, in some aspects, the functionality described herein may be provided within dedicated software modules or hardware modules configured as described herein. Also, the techniques could be fully implemented in one or more circuits or logic elements. A general purpose processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, e.g., a combination of an SPU/MPU and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with an SPU core, MPU core, or any other such configuration.

Overview of Discussion

Discussion begins with a description of an example retinal projection display system. Example operations of a retinal projection display system are then described.

Embodiments herein provide a retinal projection display system including a light source for projecting an image, a scanning mirror having a field of view larger than the image, and a reflective surface on which the image is projected, wherein the reflective surface is larger than the image. The scanning mirror projects the image onto a viewable region of the reflective surface such that the image is projected into a retina of a user. In some embodiments, the reflective surface is at least partially transparent. In some embodiments, the retinal projection display further includes an eyeglasses frame configured to be worn by the user and at least one lens mounted in the eyeglasses frame, where the reflective surface is positioned on at least a portion of the at least one lens.

In some embodiments, a pupillary distance alignment is determined during a calibration operation for the user, wherein the pupillary distance alignment identifies the viewable region of the reflective surface for a known gaze direction of the user. In some embodiments, the image is displayed at multiple locations of the reflective surface during the calibration operation, and the pupillary distance alignment is determined responsive to feedback from the user identifying the viewable region of the reflective surface.

In some embodiments, the retinal projection display system also includes a gaze tracker for tracking a gaze direction of the user, wherein the viewable region corresponds to the gaze direction. The scanning mirror is configured to dynamically move the image on the reflective surface according to the gaze direction of the user and the pupillary distance alignment of the user. In some embodiments, the scanning mirror is configured to dynamically move the image on the reflective surface according to the gaze direction of the user responsive to the gaze direction satisfying a movement threshold. In some embodiments, the wherein the scanning mirror is configured to dynamically move the image on the reflective surface according to the gaze direction of the user after a predetermined time delay. In some embodiments, the

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image is smaller than the viewable region such that scanning mirror is configured to dynamically move the image on the reflective surface according to the gaze direction of the user responsive to image moving outside of the viewable region.

In some embodiments, a scanning range of the scanning mirror is dynamically adjusted to correspond to a size of the image in the viewable region. In some embodiments, a scanning range of the scanning mirror corresponds to a size of a display area of the reflective surface, such that the light source is activated for displaying the image only when the scanning mirror is projecting the image in the viewable region.

Other embodiments described herein provide a method of retinal projection. An image from a light source is projected onto a reflective surface using a scanning mirror having a field of view larger than the image, where the reflective surface is larger than the image. In some embodiments, a scanning range of the scanning mirror is dynamically adjusted to correspond to a size of the image in the viewable region. In some embodiments, a scanning range of the scanning mirror corresponds to a size of display area of the reflective surface, such that the light source is activated for displaying the image only when the scanning mirror is projecting the image in the viewable region. In some embodiments, the reflective surface is at least partially transparent.

A viewable region of the reflective surface for a user is determined. In some embodiments, a pupillary distance alignment for the user is retrieved (e.g., from memory), wherein the pupillary distance alignment identifies the viewable region of the reflective surface for a known gaze direction of the user. In some embodiments, a gaze direction of the user is tracked, wherein the viewable region corresponds to the gaze direction.

The image is directed onto the viewable region of the reflective surface such that the image is projected into a retina of the user. In some embodiments, the image is dynamically moved on the reflective surface according to the gaze direction of the user and the pupillary distance alignment of the user using the scanning mirror.

In some embodiments, an amount of movement of the gaze direction is determined based on tracking the gaze direction of the user and, responsive to the amount of movement of the gaze direction satisfying a movement threshold, the image is moved on the reflective surface according to the gaze direction of the user and the pupillary distance alignment of the user. In some embodiments, the image is moved on the reflective surface according to the gaze direction of the user and the pupillary distance alignment of the user after a predetermined time delay. In some embodiments, responsive to determining that the image is outside of the viewable region, the image is moved on the reflective surface according to the gaze direction of the user and the pupillary distance alignment of the user.

Example Retinal Projection Display System

FIG. 1 illustrates an example retinal projection display system 100, according to some embodiments. Retinal projection display system 100 includes light source 110, scanning mirror 120, and reflective surface 130. In the illustrated embodiment, the components of retinal projection display system 100 are comprised within eyeglasses frame 160 that is configured to be worn by a user. Eyeglasses frame 160 includes at least one lens 162 mounted therein, where reflective surface 130 is positioned on at least a portion of lens 162 such that reflective surface 130 is within view of the user when gazing through lens 162. In some embodiments, reflective surface 130 is at least partially transparent, allow-

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ing the user to view through reflective surface 130 and lens 162. It should be appreciated that various components of retinal projection display system 100, such as light source 110 and scanning mirror 120, can be disposed on or within cavities eyeglasses frame 160 (e.g., within a cavity 115 of an arm/temple) and positioned such that light beam 112 is projected onto reflective surface 130.

During operation, light source 110 (e.g., a laser) projects an image onto a portion of reflective surface 130 by generating light beam 112 that is projected onto scanning mirror 120. In some embodiments, light source 110 is a single light source capable of projecting a complete image. In some embodiments, light source 110 include multiple light sources such as separate red, green, and blue (RGB) lasers that operate in coordination to project a complete image. It should be appreciated that many types of light sources can be used in accordance with the described embodiments.

Scanning mirror 120 is configured to move and direct light beam 112 such that it is scanned over reflective surface 130 to place each point of the image onto reflective surface 130, which directs light beam 112 through the user's pupil 154 of eye 150 and onto retina 152. It should be appreciated that a variety of scanning patterns can be used, as described below. It should be appreciated that the image scanning process is performed at a scanning rate fast enough (e.g., greater than 60 Hz) such that the user perceives the entire image, or as a continuous video of images. In some embodiments, scanning mirror 120 is a microelectromechanical (MEMS) device.

Scanning mirror 120 has a field of view (FOV) larger than the size of the intended viewable image and reflective surface 130 into which the viewable image is projected is also larger than the intended viewable image. Scanning mirror 120 projects the image onto a viewable region of reflective surface 130 such that the image is projected onto retina 152 of the user. The larger FOV allows for retinal projection display system 100 to properly project the image into pupil 154 and onto retina 152 independent of the movement and rotation of eye 150. In accordance with some embodiments, retinal projection display system 100 facilitate projecting the intended viewable image to align with pupil 154 by projecting over a viewable region of reflective surface 130 over a window of scanning mirror dynamic range.

A pupillary distance alignment is used to direct light beam 112 into pupil 154 of eye 150, where the pupillary distance alignment identifies the viewable region of reflective surface 130 for a known gaze direction of the user. In some embodiments, the pupillary distance alignment is determined during a calibration operation for the user. In some embodiments, the image is displayed at multiple locations of reflective surface 130 during the calibration operation, and the pupillary distance alignment is determined responsive to feedback from the user identifying the viewable region of the reflective surface. For example, the user feedback can be provided using a user interface of retinal projection display system 100, and can be received in many ways, e.g., voice commands, buttons located on eyeglasses frame 160, an application on a connected device such as a smart phone, etc.

In some embodiments, retinal projection display system 100 also includes gaze tracker 140 for tracking a gaze direction of the user. The viewable region of reflective surface 130 corresponds to the gaze direction of the user. Scanning mirror 120 is configured to dynamically move the image on reflective surface 130 according to the gaze direction of the user and the pupillary distance alignment of the user. Since the pupillary distance alignment for the user

identifies the viewable region of reflective surface **130** for a known gaze direction of the user, scanning mirror **120** can move the image to correspond to the viewable region of reflective surface according to the gaze direction.

In some embodiments, to avoid jitter of the viewable image, scanning mirror **120** is configured to dynamically move the image on reflective surface **130** according to the gaze direction of the user responsive to the gaze direction satisfying a movement threshold. For instance, scanning mirror **120** only moves the image on reflective surface if sufficient movement of the gaze direction is detected. In some embodiments, jitter is accounted for by providing a rendered image smaller than the viewable region of reflective surface **130** such that scanning mirror **120** is configured to dynamically move the image on reflective surface **130** according to the gaze direction of the user responsive to image moving outside of the viewable region. This allows the image to be viewed over a larger range of positions on reflective surface **130** and minimizes jitter.

In some embodiments, to avoid image smearing, scanning mirror **120** is configured to dynamically move the image on reflective surface **130** according to the gaze direction of the user after a predetermined time delay after the change in gaze direction, allowing eye **150** to settle in the new gaze direction prior to moving the image.

In some embodiments, a scanning range of scanning mirror **120** is dynamically adjusted to correspond to a size of the image in the viewable region. In other embodiments, a scanning range of scanning mirror **120** corresponds to a size of a display area of reflective surface **130**, such that light source **110** is activated for displaying the image only when scanning mirror **120** is projecting the image in the viewable region of reflective surface **130**.

FIG. 2 illustrates a functional block diagram of an example retinal projection display system **200**, according to some embodiments. Retinal projection display system **200** includes light source **210**, scanning mirror **220**, gaze tracker **230**, and reflective surface. Light source **210** receives image data **205** from a data source for projection. It should be appreciated that image data **205** can include any type of data for displaying or rendering an image, including static image data, video data (e.g., a series of images), or other data for visualization by a user. In some embodiments, light source **210** is a single light source capable of projecting a complete image. In some embodiments, light source **210** include multiple light sources such as separate red, green, and blue (RGB) lasers that operate in coordination to project a complete image. It should be appreciated that many types of light sources can be used in accordance with the described embodiments.

Light source **210** (e.g., a laser) projects image **215** onto scanning mirror **220**. It should be appreciated that image **215** is projected as a scan of pixels of image **215**, where scanning mirror **220** dynamically moves to position each pixel at the proper location of reflective surface **240** for rendering. Scanning mirror **220** is configured to move and direct pixels of image **215** such that they are scanned over reflective surface **240** to place each point of the image onto reflective surface **240**, which directs image **215** into the user's pupil and onto their retina. It should be appreciated that a variety of scanning patterns can be used, as described below. It should be appreciated that the image scanning process is performed at a scanning rate fast enough (e.g., greater than 60 Hz) such that the user perceives the entire image, or as a continuous video of images.

Scanning mirror **220** utilizes pupillary distance alignment **225** for the user and gaze direction **235** to control the

position of pixels of image **215** such that they are directed onto the user's retina. The pupillary distance alignment identifies the viewable region of reflective surface **240** for a known gaze direction of the user.

FIG. 3 illustrates an example pupillary distance alignment operation **300** by projecting an alignment image onto a reflective surface, according to some embodiments. In some embodiments, pupillary distance alignment operation **300** is determined during a calibration operation for the user. In some embodiments, during pupillary distance alignment operation **300**, alignment image **310** is displayed at multiple locations of reflective surface **330**, and the pupillary distance alignment is determined responsive to feedback from the user identifying viewable region **320** of reflective surface **330**. For example, the user feedback can be provided using a user interface of retinal projection display system **100**, and can be received in many ways, e.g., voice commands, buttons located on eyeglasses frame **160**, an application on a connected device such as a smart phone, etc.

During pupillary distance alignment operation **300**, the user is instructed (e.g., via a user interface) to look in a particular direction (e.g., straight ahead). Alignment image **310** is rendered on reflective surface **330** and moved over the dynamic range of the scanning mirror to display alignment image at multiple locations on reflective surface **330**. The user provides feedback (e.g., when prompted) as to whether alignment image **310** is fully visible, partially visible, or not visible. When alignment image **310** is within viewable region **320** and is visible to the user, e.g., partially or fully, the user provides feedback to indicate that alignment image **310** is visible. In some embodiments, alignment image **310** is adapted to help with alignment. For example, alignment image **310** may include information identifying portions of alignment image **310** such as characters, arrows, colors, or other indicators, that the user can use to indicate which part of alignment image **310** they see so that the retinal projection display system knows how to move alignment image **310** into viewable region **320**.

As illustrated, alignment image **310** is projected onto reflective surface **330**. In the illustrated example, at first time **340**, alignment image **310** is not within the user's viewable region **320** of reflective surface **330** corresponding to the user's known gaze direction (e.g., straight forward). The user provides feedback that alignment image **310** is not visible to the user. At second time **350**, alignment image **310** is moved to a different location of reflective surface **330** that is still not within viewable region **320**. As illustrated, viewable region **320** is substantially static during pupillary distance alignment operation **300**. The user provides feedback that alignment image **310** is not visible to the user.

At third time **360**, alignment image **310** is moved to a different location of reflective surface **330** that is partially within viewable region **320**. The user provides feedback that alignment image **310** is partially visible to the user. At fourth time **370**, alignment image **310** is moved to a different location of reflective surface **330** that is fully within viewable region **320**. The user provides feedback that alignment image **310** is fully visible to the user. The position of alignment image **310** at fourth time **370** is stored and used as the pupillary distance alignment for the user's known gaze direction (e.g., straight forward). The pupillary distance alignment is stored (e.g., within memory of the retinal projection display system).

With reference to FIG. 2, gaze tracker **230** is for tracking gaze direction **235** of the user. The viewable region of reflective surface **240** corresponds to the gaze direction of the user such that as gaze direction **235** moves, the viewable

region of the user on reflective surface **240** moves as well. Scanning mirror **220** is configured to dynamically move image **215** on reflective surface **240** according to gaze direction **235** of the user and pupillary distance alignment **225** of the user. Since pupillary distance alignment **225** for the user identifies the viewable region of reflective surface **240** for a known gaze direction of the user, scanning mirror **220** can move image **215** to correspond to the viewable region of reflective surface **240** according to gaze direction **235**.

FIG. **4** illustrates an example gaze tracking operation **400** used during projection of an image onto a reflective surface, according to some embodiments. During gaze tracking operation **400**, a gaze tracker (e.g., gaze tracker **140** of FIG. **1** or gaze tracker **230** of FIG. **2**) is configured to track the gaze direction of the user. It should be appreciated that gaze tracking is understood by one of ordinary skill in the art, and that any type of gaze tracking technology or operation can be utilized or executed to perform gaze tracking operation **400**. As the pupillary distance alignment identifying the viewable region for a known gaze direction of the user is known, the gaze direction can be used to move the viewable region of the user for any gaze direction.

As illustrated, an image is projected onto viewable region **420** of reflective surface **410**. In the illustrated example, at first time **430**, the image is within the user's viewable region **420** of reflective surface **410** corresponding to the user's gaze direction as identified by the gaze tracking. At second time **440**, where the user's gaze direction has moved, viewable region **420** is moved, as is the image, corresponding to the user's gaze direction as identified by the gaze tracking at the later time.

In some embodiments, to avoid jitter of the viewable image, viewable region **420** is only moved if sufficient movement of the gaze direction is detected (e.g., a movement threshold is satisfied). In some embodiments, jitter is accounted for by providing a rendered image smaller than viewable region **420**, such that viewable region **420** is moved responsive to the gaze direction of the user moving outside of viewable region **420**. This allows the image to be viewed over a larger range of positions and minimizes jitter. In some embodiments, to avoid image smearing, viewable region **420** is moved according to the gaze direction of the user after a predetermined time delay after the change in gaze direction, allowing the user's eye to settle in the new gaze direction prior to moving the image.

With reference to FIG. **2**, scanning mirror **220** projects image **215** (e.g., pixel by pixel) onto a viewable region of reflective surface **240** such that image **215** is projected into a retina of a user. In some embodiments, a scanning range of scanning mirror **220** is dynamically adjusted to correspond to a size of the image in the viewable region. In other embodiments, a scanning range of scanning mirror **220** corresponds to a size of a display area of reflective surface **240**, such that light source **210** is activated for displaying the image only when scanning mirror **220** is projecting the image in the viewable region of reflective surface **240**.

FIGS. **5A** and **5B** illustrates an example image rendering operation **500** where the scanning range of the scanning mirror is the size of the projected image, according to an embodiment. FIG. **5A** illustrates an example scanning pattern over the full scanning range **510** of the scanning mirror, where the scanning mirror is capable of rendering an image anywhere on the reflective surface, where the reflective surface is larger than a viewable region of the scanning surface. As illustrated, the scanning mirror is configured to move in the x and y directions, with center position **520**

being the identified gaze direction of the user and the center position of the viewable region. Eye tracking is used to identify center position of the viewable region.

FIG. **5B** illustrates an example scanning pattern where the scanning range of the scanning mirror is the size of the projected image. As illustrated in FIG. **5B**, center position **530** is identified using gaze tracking, where center position **530** is moved relative to center position **520**. The scanning mirror panning angles are controlled so that the x and y scanning range **550** covers the viewable region **540** as identified by center position B. In the illustrated embodiment, scanning range **550** of the scanning mirror is reduced to the cover viewable region **540** of the reflective surface rather than the entire reflective surface.

FIG. **6** illustrates an example image rendering operation **600** where the scanning range **610** of the scanning mirror is larger than the size of the projected image, according to an embodiment. As illustrated, the scanning mirror is configured to move in the x and y directions, with center position **620** being the identified gaze direction of the user and the center position of the viewable region. Eye tracking is used to identify center position of the viewable region.

As illustrated in FIG. **6**, center position **630** is identified using gaze tracking, where center position **630** is moved relative to center position **620**. Depending on gaze direction according to center position **630**, the light source is only activated when the scanning mirror is within viewable region **640**. The scanning mirror panning angles are controlled so that the x and y scanning range **610** covers the entire reflective surface area, but only activates the light source when the scanning mirror is within viewable region **640**.

FIG. **7** is a block diagram of an example electronic device **700** upon which embodiments of the present invention can be implemented. FIG. **7** illustrates one example of a type of electronic device **700** (e.g., a computer system) that can be used in accordance with or to implement various embodiments which are discussed herein. It should be appreciated that embodiments of the described retinal projection display system can be implemented using example electronic device **700**.

It is appreciated that electronic device **700** of FIG. **7** is only an example and that embodiments as described herein can operate on or within a number of different computer systems including, but not limited to, general purpose networked computer systems, embedded computer systems, mobile electronic devices, smart phones, server devices, client devices, various intermediate devices/nodes, stand-alone computer systems, media centers, handheld computer systems, multi-media devices, and the like. In some embodiments, electronic device **700** of FIG. **7** is well adapted to having peripheral tangible computer-readable storage media **702** such as, for example, an electronic flash memory data storage device, a floppy disc, a compact disc, digital versatile disc, other disc based storage, universal serial bus "thumb" drive, removable memory card, and the like coupled thereto. The tangible computer-readable storage media is non-transitory in nature.

Electronic device **700** of FIG. **7** includes an address/data bus **704** for communicating information, and a processor **706A** coupled with bus **704** for processing information and instructions. Bus **704** may be any suitable bus or interface to include, without limitation, a peripheral component interconnect express (PCIe) bus, a universal serial bus (USB), a universal asynchronous receiver/transmitter (UART) serial bus, a suitable advanced microcontroller bus architecture (AMBA) interface, an Inter-Integrated Circuit (I2C) bus, a

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serial digital input output (SDIO) bus, a serial peripheral interface (SPI) or other equivalent.

As depicted in FIG. 7, electronic device 700 is also well suited to a multi-processor environment in which a plurality of processors 706A, 706B, and 706C are present. Conversely, electronic device 700 is also well suited to having a single processor such as, for example, processor 706A. Processors 706A, 706B, and 706C may be any of various types of microprocessors. Electronic device 700 also includes data storage features such as a computer usable volatile memory 708, e.g., random access memory (RAM), coupled with bus 704 for storing information and instructions for processors 706A, 706B, and 706C. Electronic device 700 also includes computer usable non-volatile memory 710, e.g., read only memory (ROM), coupled with bus 704 for storing static information and instructions for processors 706A, 706B, and 706C. Also present in electronic device 700 is a data storage unit 712 (e.g., a magnetic or optical disc and disc drive) coupled with bus 704 for storing information and instructions. Electronic device 700 also includes an alphanumeric input device 714 including alphanumeric and function keys coupled with bus 704 for communicating information and command selections to processor 706A or processors 706A, 706B, and 706C. Electronic device 700 also includes a cursor control device 716 coupled with bus 704 for communicating user input information and command selections to processor 706A or processors 706A, 706B, and 706C. In one embodiment, electronic device 700 also includes a display device 718 coupled with bus 704 for displaying information. Depending on the architecture, different bus configurations may be employed as desired. For example, additional buses may be used to couple the various components of electronic device 700, such as by using a dedicated bus between processor 706A and memory computer usable volatile memory 708 or computer usable non-volatile memory 710.

Referring still to FIG. 7, display device 718 of FIG. 7 may include a light source (e.g., light source 110 of FIG. 1) for projecting image data onto a reflective surface. In other embodiments, display device 718 may be a liquid crystal device (LCD), light emitting diode display (LED) device, plasma display device, a touch screen device, or other display device suitable for creating graphic images and alphanumeric characters recognizable to a user. Cursor control device 716 allows the computer user to dynamically signal the movement of a visible symbol (cursor) on a display screen of display device 718 and indicate user selections of selectable items displayed on display device 718. Many implementations of cursor control device 716 are known in the art including a trackball, mouse, touch pad, touch screen, joystick or special keys on alphanumeric input device 714 capable of signaling movement of a given direction or manner of displacement. Alternatively, it will be appreciated that a cursor can be directed and/or activated via input from alphanumeric input device 714 using special keys and key sequence commands. Electronic device 700 is also well suited to having a cursor directed by other means such as, for example, voice commands. In various embodiments, alphanumeric input device 714, cursor control device 716, and display device 718, or any combination thereof (e.g., user interface selection devices), may collectively operate to provide a graphical user interface (GUI) 730 under the direction of a processor (e.g., processor 706A or processors 706A, 706B, and 706C). GUI 730 allows user to interact with electronic device 700 through graphical representations presented on display device 718 by interacting with alphanumeric input device 714 and/or cursor control device 716.

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Electronic device 700 also includes an I/O device 720 for coupling electronic device 700 with external entities. For example, in one embodiment, I/O device 720 is a modem for enabling wired or wireless communications between electronic device 700 and an external network such as, but not limited to, the Internet. In one embodiment, I/O device 720 includes a transmitter. Electronic device 700 may communicate with a network by transmitting data via I/O device 720.

Referring still to FIG. 7, various other components are depicted for electronic device 700. Specifically, when present, an operating system 722, applications 724, modules 726, and data 728 are shown as typically residing in one or some combination of computer usable volatile memory 708 (e.g., RAM), computer usable non-volatile memory 710 (e.g., ROM), and data storage unit 712. In some embodiments, all or portions of various embodiments described herein are stored, for example, as an application 724 and/or module 726 in memory locations within RAM 708, computer-readable storage media within data storage unit 712, peripheral computer-readable storage media 702, and/or other tangible computer-readable storage media. Example Operations for Operating a Retinal Projection Display System

FIG. 8 illustrates an example process of retinal projection and FIG. 9 illustrates an example process for determining a pupillary distance alignment, according to some embodiments. Procedures of these methods will be described with reference to elements and/or components of various figures described herein. It is appreciated that in some embodiments, the procedures may be performed in a different order than described, that some of the described procedures may not be performed, and/or that one or more additional procedures to those described may be performed. The flow diagrams include some procedures that, in various embodiments, are carried out by one or more processors (e.g., a host processor or a sensor processor) under the control of computer-readable and computer-executable instructions that are stored on non-transitory computer-readable storage media. It is further appreciated that one or more procedures described in the flow diagrams may be implemented in hardware, or a combination of hardware with firmware and/or software.

With reference to FIG. 8, flow diagram 800 illustrates an example process of retinal projection, according to some embodiments. At procedure 810 of flow diagram 800, an image from a light source is projected onto a reflective surface using a scanning mirror having a field of view larger than the image, where the reflective surface is larger than the image. In some embodiments, a scanning range of the scanning mirror is dynamically adjusted to correspond to a size of the image in the viewable region. In some embodiments, a scanning range of the scanning mirror corresponds to a size of display area of the reflective surface, such that the light source is activated for displaying the image only when the scanning mirror is projecting the image in the viewable region. In some embodiments, the reflective surface is at least partially transparent.

At procedure 820, a viewable region of the reflective surface for a user is determined. In some embodiments, as shown at procedure 822, a pupillary distance alignment for the user is retrieved (e.g., from memory), wherein the pupillary distance alignment identifies the viewable region of the reflective surface for a known gaze direction of the user. In some embodiments, as shown at procedure 824, a gaze direction of the user is tracked, wherein the viewable region corresponds to the gaze direction.

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At procedure **830**, the image is directed onto the viewable region of the reflective surface such that the image is projected into a retina of the user. In some embodiments, as shown at procedure **832**, the image is dynamically moved on the reflective surface according to the gaze direction of the user and the pupillary distance alignment of the user using the scanning mirror.

In some embodiments, as shown at procedure **834**, an amount of movement of the gaze direction is determined based on tracking the gaze direction of the user and, responsive to the amount of movement of the gaze direction satisfying a movement threshold, the image is moved on the reflective surface according to the gaze direction of the user and the pupillary distance alignment of the user. In some embodiments, as shown at procedure **836**, the image is moved on the reflective surface according to the gaze direction of the user and the pupillary distance alignment of the user after a predetermined time delay. In some embodiments, as shown at procedure **838**, responsive to determining that the image is outside of the viewable region, the image is moved on the reflective surface according to the gaze direction of the user and the pupillary distance alignment of the user.

FIG. **9** illustrates an example flow diagram **900** for determining a pupillary distance alignment, e.g., during a calibration operation, according to some embodiments. At procedure **910** of flow diagram **900**, an alignment image is projected onto the reflective surface. At procedure **920**, user feedback is received regarding the viewability of the alignment image while the user is gazing in a known gaze direction. At procedure **930**, it is determined whether the alignment image is in the viewable region according to the user feedback. If the alignment image is not in the viewable region, flow diagram **900** proceeds to procedure **940**. At procedure **940**, the position of the alignment image is moved on the reflective surface to another position. If the alignment image is within in the viewable region, flow diagram **900** proceeds to procedure **950**.

At procedure **950**, it is determined whether the alignment image is completely within the viewable region according to the user feedback. If the alignment image is not completely within the viewable region, flow diagram **900** proceeds to procedure **940**. At procedure **940**, the position of the alignment image is moved on the reflective surface to another position. If the alignment image is completely within in the viewable region, flow diagram **900** proceeds to procedure **960**. At procedure **960**, the pupillary distance alignment identifying the viewable region of the reflective surface for the known gaze direction of the user is determined. In some embodiments, the pupillary distance alignment is stored (e.g., in memory) for retrieval during retinal projection display operation.

CONCLUSION

The examples set forth herein were presented in order to best explain, to describe particular applications, and to thereby enable those skilled in the art to make and use embodiments of the described examples. However, those skilled in the art will recognize that the foregoing description and examples have been presented for the purposes of illustration and example only. Many aspects of the different example embodiments that are described above can be combined into new embodiments. The description as set forth is not intended to be exhaustive or to limit the embodiments to the precise form disclosed. Rather, the

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specific features and acts described above are disclosed as example forms of implementing the claims.

Reference throughout this document to “one embodiment,” “certain embodiments,” “an embodiment,” “various embodiments,” “some embodiments,” or similar term means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, the appearances of such phrases in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics of any embodiment may be combined in any suitable manner with one or more other features, structures, or characteristics of one or more other embodiments without limitation.

What is claimed is:

1. A retinal projection display system comprising:
 - a light source for projecting an image;
 - a scanning mirror having a field of view larger than the image; and
 - a reflective surface on which the image is projected, wherein the reflective surface is larger than the image; wherein the scanning mirror projects the image onto a viewable region of the reflective surface such that the image is projected into a retina of a user, wherein a pupillary distance alignment is determined during a calibration operation for the user, wherein the pupillary distance alignment identifies the viewable region of the reflective surface for a known gaze direction of the user, wherein the image is displayed at multiple locations of the reflective surface during the calibration operation, and the pupillary distance alignment is determined responsive to feedback from the user identifying the viewable region of the reflective surface.
2. The retinal projection display system of claim 1, further comprising:
 - a gaze tracker for tracking a gaze direction of the user, wherein the viewable region corresponds to the gaze direction;
 - wherein the scanning mirror is configured to dynamically move the image on the reflective surface according to the gaze direction of the user and the pupillary distance alignment of the user.
3. The retinal projection display system of claim 2, wherein the scanning mirror is configured to dynamically move the image on the reflective surface according to the gaze direction of the user responsive to the gaze direction satisfying a movement threshold.
4. The retinal projection display system of claim 2, wherein the scanning mirror is configured to dynamically move the image on the reflective surface according to the gaze direction of the user after a predetermined time delay.
5. The retinal projection display system of claim 2, wherein the image is smaller than the viewable region such that scanning mirror is configured to dynamically move the image on the reflective surface according to the gaze direction of the user responsive to image moving outside of the viewable region.
6. The retinal projection display system of claim 1, wherein the reflective surface is at least partially transparent.
7. The retinal projection display system of claim 1, further comprising:
 - an eyeglasses frame configured to be worn by the user; and
 - at least one lens mounted in the eyeglasses frame, where the reflective surface is positioned on at least a portion of the at least one lens.

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8. A retinal projection display system comprising:
 a light source for projecting an image;
 a scanning mirror having a field of view larger than the image; and
 a reflective surface on which the image is projected, wherein the reflective surface is larger than the image; wherein the scanning mirror projects the image onto a viewable region of the reflective surface such that the image is projected into a retina of a user, and wherein a scanning range of the scanning mirror is dynamically adjusted to correspond to a size of the image in the viewable region.
9. A retinal projection display system comprising:
 a light source for projecting an image;
 a scanning mirror having a field of view larger than the image; and
 a reflective surface on which the image is projected, wherein the reflective surface is larger than the image; wherein the scanning mirror projects the image onto a viewable region of the reflective surface such that the image is projected into a retina of a user, and wherein a scanning range of the scanning mirror corresponds to a size of a display area of the reflective surface, such that the light source is activated for displaying the image only when the scanning mirror is projecting the image in the viewable region.
10. A method of retinal projection, the method comprising:
 projecting an image from a light source onto a reflective surface using a scanning mirror having a field of view larger than the image, wherein the reflective surface is larger than the image;
 determining a viewable region of the reflective surface for a user, wherein the determining the viewable region of the reflective surface of the user comprises:
 retrieving a pupillary distance alignment for the user, wherein the pupillary distance alignment identifies the viewable region of the reflective surface for a known gaze direction of the user, wherein the pupillary distance alignment is determined during a calibration operation for the user, the method further comprising:
 during the calibration operation:
 projecting an alignment image onto the reflective surface;
 moving the alignment image across a display area of the reflective surface; and
 responsive to feedback from the user, determining the pupillary distance alignment identifying the viewable region of the reflective surface for the known gaze direction of the user; and
 directing the image onto the viewable region of the reflective surface such that the image is projected into a retina of the user.
11. The method of claim 10, wherein the determining the viewable region of the reflective surface of the user further comprises:
 tracking a gaze direction of the user, wherein the viewable region corresponds to the gaze direction.
12. The method of claim 11, wherein the directing the image onto the viewable region of the reflective surface such that the image is projected into a retina of the user comprises:

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- dynamically moving the image on the reflective surface according to the gaze direction of the user and the pupillary distance alignment of the user using the scanning mirror.
13. The method of claim 12, wherein the dynamically moving the image on the reflective surface according to the gaze direction of the user and the pupillary distance alignment of the user using the scanning mirror comprises:
 determining an amount of movement of the gaze direction based on the tracking the gaze direction of the user; and
 responsive to the amount of movement of the gaze direction satisfying a movement threshold, moving the image on the reflective surface according to the gaze direction of the user and the pupillary distance alignment of the user.
14. The method of claim 12, wherein the dynamically moving the image on the reflective surface according to the gaze direction of the user and the pupillary distance alignment of the user using the scanning mirror comprises:
 moving the image on the reflective surface according to the gaze direction of the user and the pupillary distance alignment of the user after a predetermined time delay.
15. The method of claim 12, wherein the image is smaller than the viewable region and wherein the dynamically moving the image on the reflective surface according to the gaze direction of the user and the pupillary distance alignment of the user using the scanning mirror comprises:
 determining whether the image is outside of the viewable region; and
 responsive to determining that the image is outside of the viewable region, moving the image on the reflective surface according to the gaze direction of the user and the pupillary distance alignment of the user.
16. The method of claim 10, wherein the reflective surface is at least partially transparent.
17. A method of retinal projection, the method comprising:
 projecting an image from a light source onto a reflective surface using a scanning mirror having a field of view larger than the image, wherein the reflective surface is larger than the image;
 determining a viewable region of the reflective surface for a user;
 directing the image onto the viewable region of the reflective surface such that the image is projected into a retina of the user; and
 wherein a scanning range of the scanning mirror is dynamically adjusted to correspond to a size of the image in the viewable region.
18. A method of retinal projection, the method comprising:
 projecting an image from a light source onto a reflective surface using a scanning mirror having a field of view larger than the image, wherein the reflective surface is larger than the image;
 determining a viewable region of the reflective surface for a user;
 directing the image onto the viewable region of the reflective surface such that the image is projected into a retina of the user; and
 wherein a scanning range of the scanning mirror corresponds to a size of display area of the reflective surface, such that the light source is activated for displaying the image only when the scanning mirror is projecting the image in the viewable region.

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